

Customer Engagement and Product Utilization Analytics for Retention Strategy

A Data-Driven Study of a European Retail Bank's Customer Base

Analysis of 10,000 customer records covering demographics, account engagement, product holdings, and churn outcomes to evaluate whether behavioral engagement — rather than financial strength alone — drives customer retention.

Abstract

Banks often assume that financially strong customers — those with high balances or salaries — are inherently loyal, yet behavioral engagement may be a stronger predictor of retention. This paper presents an exploratory analysis of 10,000 customer records from a European retail bank operating in France, Germany, and Spain. We examine the relationship between account activity, product holdings, and churn (Exited), using cross-tabulation and segment-level churn-rate analysis. We find an overall churn rate of 20.4%, that inactive members churn at nearly double the rate of active members (26.9% vs. 14.3%), that customers holding two products churn at just 7.6% versus 27.7% for single-product holders, and that high-balance customers who are also disengaged reach a 32.3% churn rate — the highest risk segment identified. Germany shows a materially elevated churn rate (32.4%) relative to France and Spain (~16%). We conclude with recommendations for engagement-driven retention and product-bundling strategy.

1. Introduction

Customer behavior and engagement — not just demographics — increasingly determine long-term retention at retail banks. A customer may appear financially strong on paper (high balance, high salary) yet still churn due to low engagement, limited product adoption, or weak relationship depth with the institution. Conversely, banks that understand how customers actually use their products and services are better positioned to design effective cross-sell strategies, loyalty programs, and engagement-driven retention initiatives.

This project evaluates retention through the lens of customer behavior and relationship strength, rather than relying solely on demographic or financial-strength indicators.

1.1 Problem Statement

Despite having data on customer engagement and product usage, banks often lack quantitative insight into which behaviors drive retention, clarity on whether product depth reduces churn, and evidence on whether high balances alone ensure loyalty. As a result, retention strategies are frequently generic and misaligned with actual customer behavior.

1.2 Objectives

- Evaluate the relationship between engagement (activity status) and churn
- Measure the retention impact of product count and product mix
- Identify disengaged yet high-value (high-balance) customers
- Support engagement-driven retention strategies and improve product-bundling decisions

2. Data and Methodology

2.1 Dataset

The dataset comprises 10,000 customer records, each describing a unique bank customer across France, Germany, and Spain. Fields include demographic attributes (age, gender, geography), financial attributes (credit score, balance, estimated salary), relationship attributes (tenure, number of products, credit card ownership, active-member status), and the binary churn outcome (Exited).

Column	Description
CustomerId / Surname	Unique identifier and customer surname
CreditScore	Customer creditworthiness
Geography	France, Spain, or Germany
Gender / Age / Tenure	Demographic and relationship-length fields
Balance	Account balance
NumOfProducts	Number of bank products held
HasCrCard / IsActiveMember	Credit card ownership; activity indicator
EstimatedSalary	Estimated annual salary
Exited	Churn indicator (target variable)

2.2 Analytical Methodology

- Data Ingestion & Validation — loaded and validated all fields; confirmed binary-variable consistency and churn-label accuracy
- Engagement Classification — segmented customers into active/inactive, and cross-referenced with balance tier
- Product Utilization Analysis — computed churn rate by number of products held
- Financial Commitment vs. Engagement Analysis — cross-analyzed balance against activity status to detect at-risk premium customers
- Retention Strength Assessment — defined a 'sticky customer' profile and measured churn stability across engagement tiers

2.3 Key Performance Indicators

KPI	Definition
Engagement Retention Ratio	Churn rate, active vs. inactive members
Product Depth Index	Churn rate by number of products held
High-Balance Disengagement Rate	Churn rate among high-balance, inactive customers
Credit Card Stickiness Score	Churn rate by credit card ownership
Relationship Strength Index	Combined engagement and product-depth churn rate

3. Findings

3.1 Portfolio Overview

Of 10,000 customers, 20.4% churned during the observation period. 51.5% of customers are classified as active members, 70.6% hold a credit card, and 50.8% hold only a single product — the modal and, as shown below, highest-risk holding level.

3.2 Engagement Drives Retention

Active members churn at less than half the rate of inactive members, confirming that behavioral engagement — not financial standing — is a primary retention lever.

Activity Status	Customers	Churn Rate
Active member	5,151	14.3%
Inactive member	4,849	26.9%

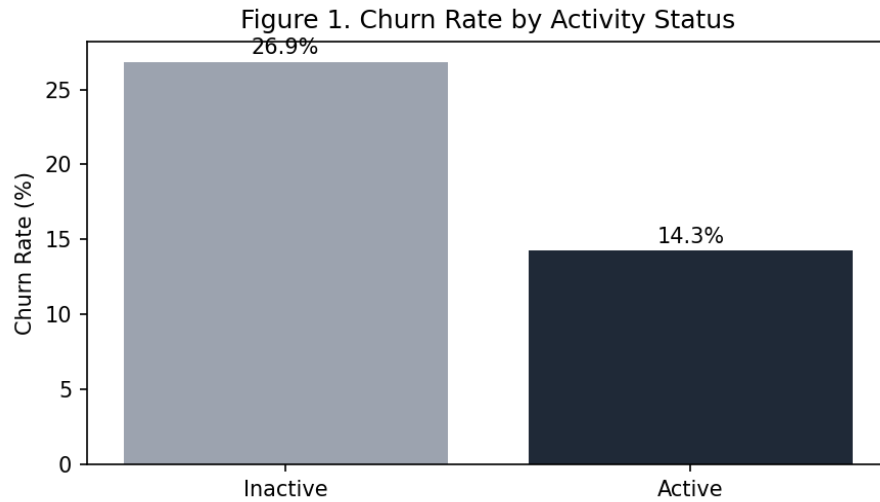


Figure 1. Inactive members churn at nearly double the rate of active members.

3.3 Product Depth Index

Churn is sharply non-linear across product count. Two-product holders are the 'sweet spot,' churning at less than a third the rate of single-product holders, while customers holding three or more products churn at an alarming rate — likely reflecting cross-sell overreach, forced bundling, or distressed accounts rather than genuine loyalty.

Products Held	Customers	Churn Rate
1	5,084	27.7%
2	4,590	7.6%
3	266	82.7%
4	60	100.0%

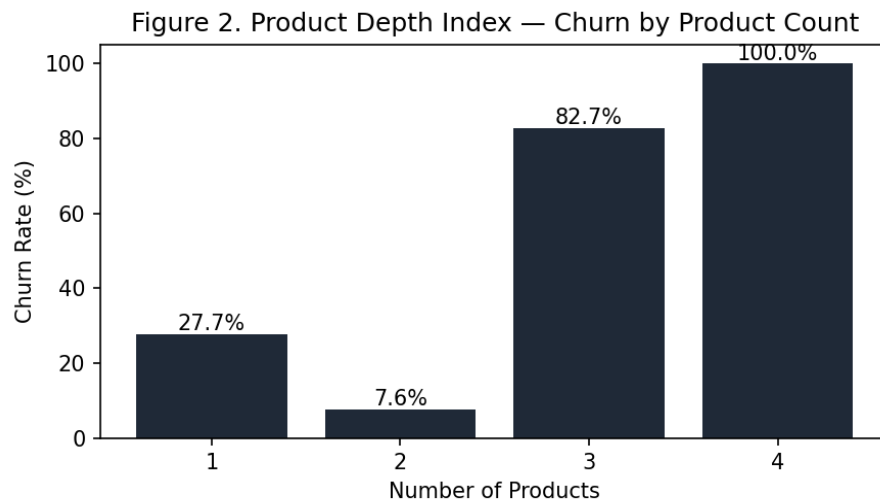


Figure 2. Two-product holders are the retention 'sweet spot'; 3–4 products signal distress.

3.4 High-Balance Disengagement Risk

Cross-referencing balance tier (above vs. below the \$97,199 median) with activity status isolates the portfolio's highest-risk segment: high-balance, inactive customers churn at 32.3% — more than three times the rate of low-balance, active customers (10.7%). This segment represents customers who appear financially valuable but are quietly disengaging, a pattern of 'silent churn' among premium customers.

Balance Tier	Active	Inactive
Below median	21.2%	10.7%
Above median	17.9%	32.3%

Note: cells show churn rate by activity status within each balance tier.

Figure 3. High-Balance Disengagement Risk

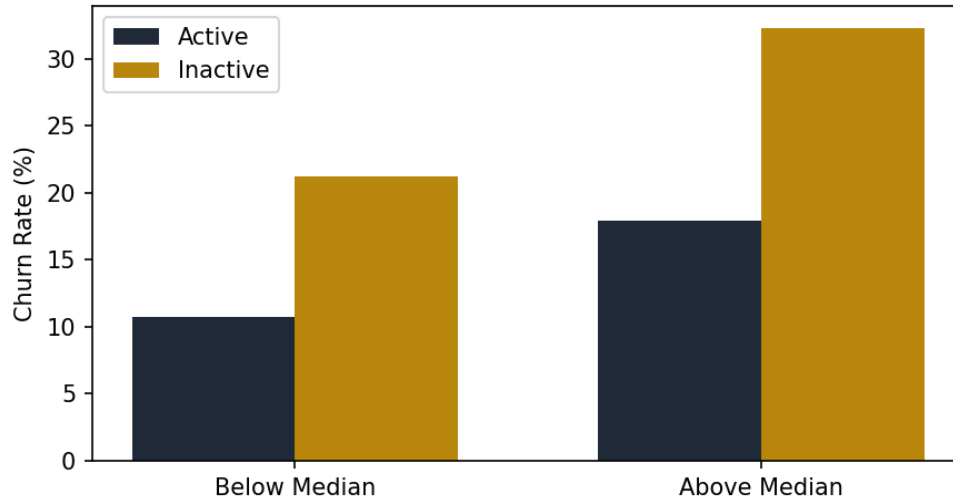


Figure 3. High-balance, inactive customers reach the highest churn rate in the portfolio.

3.5 Geographic Distribution

Germany exhibits a churn rate roughly double that of France and Spain, despite comparable product and engagement mechanics, suggesting market-specific competitive pressure or service-quality factors warranting further investigation.

Geography	Customers	Churn Rate
France	5,014	16.2%
Germany	2,509	32.4%
Spain	2,477	16.7%

Figure 4. Churn Rate by Geography

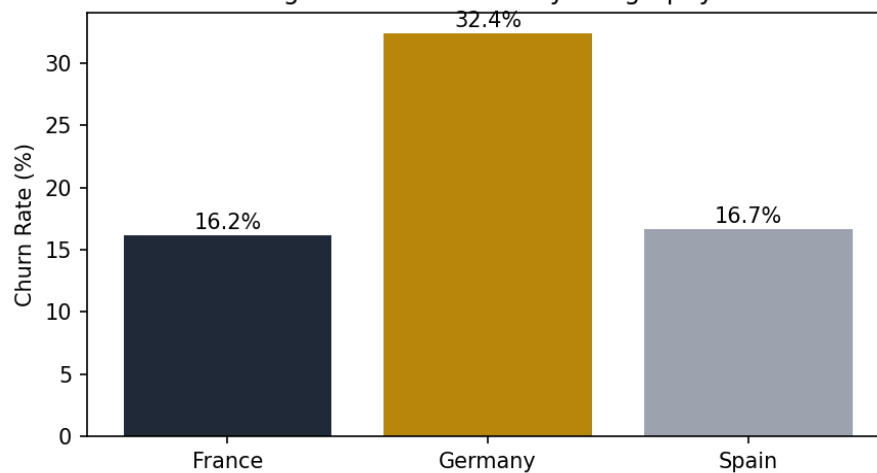


Figure 4. Germany's churn rate is roughly double that of France and Spain.

3.6 Credit Card Ownership and Age

Credit card ownership shows negligible association with churn (20.8% vs. 20.2%), indicating it is not a meaningful retention lever on its own. Age, by contrast, shows a strong relationship: churn rises from 7.5% among customers under 30 to a peak of 56.2% among 50-to-60-year-olds, before declining among the oldest cohort — a pattern consistent with mid-career life-stage disruptions (e.g., competing financial products, retirement planning shifts) rather than simple loyalty decay.

Age Band	Churn Rate
18–30	7.5%
31–40	12.1%
41–50	34.0%
51–60	56.2%
61+	24.8%

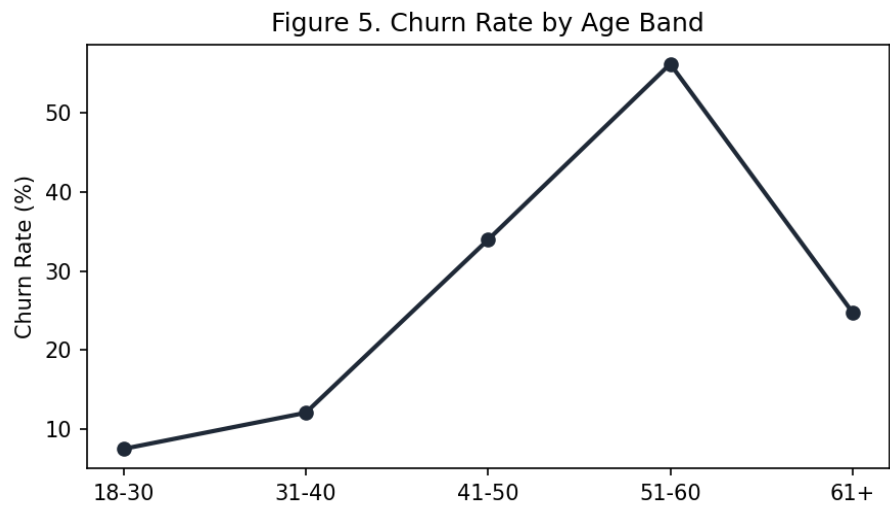


Figure 5. Churn peaks among 51–60 year-olds before declining in the oldest cohort.

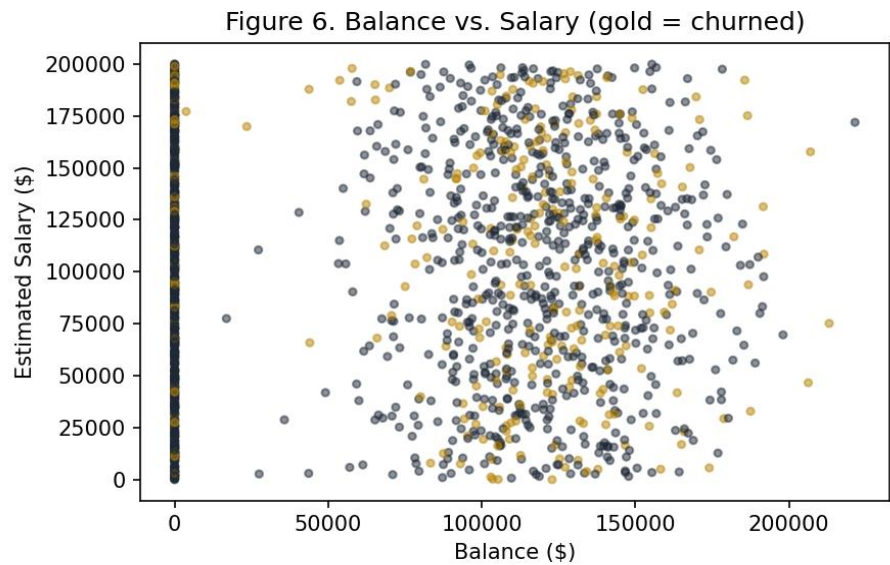


Figure 6. Balance vs. salary, sampled customers colored by churn outcome — no simple linear separation is visible, reinforcing that behavioral, not financial, variables drive churn.

3.7 The 'Sticky Customer' Profile

Customers who are both active members and hold exactly two products form a clear 'sticky' segment: 2,446 customers (24.5% of the base) churn at just 5.6% — roughly a quarter of the portfolio average — establishing this combination as the target profile for retention and cross-sell strategy.

Figure 7. "Sticky Customer" Segment Share

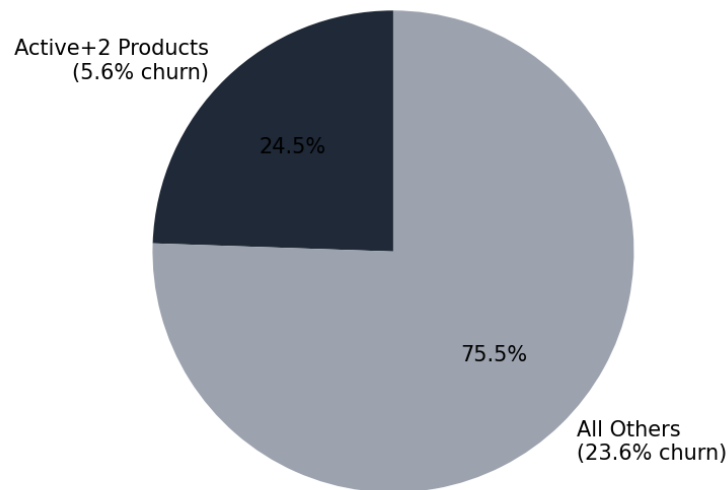


Figure 7. The 'sticky' segment (active, 2 products) churns at a quarter of the base rate.

4. Discussion

Three patterns emerge. First, engagement is a stronger and more actionable predictor of churn than financial strength alone — high balance without activity is a warning sign, not a safety net. Second, product depth has a non-monotonic relationship with retention: a second product meaningfully anchors the relationship, but a third or fourth product is associated with sharply elevated churn, likely reflecting forced or distressed multi-product holdings rather than genuine cross-sell success. Third, geographic and life-stage effects (Germany, and customers aged 41–60) suggest that retention risk is unevenly distributed and may warrant targeted, rather than uniform, intervention.

5. Recommendations

- Launch a proactive outreach program targeting high-balance, inactive customers (32.3% churn) before they silently exit
- Promote the two-product 'sticky' bundle as the core cross-sell target, rather than pursuing product count indiscriminately
- Investigate the mechanics behind elevated churn among 3–4 product holders to rule out forced bundling or service friction
- Conduct a market-specific review of German operations given churn nearly double that of France and Spain
- Design age-aware retention journeys for the 41–60 cohort, where churn peaks

- Operationalize these KPIs via the accompanying Streamlit dashboard for continuous, self-service monitoring

6. Limitations and Future Work

This analysis is cross-sectional and does not capture the sequence or timing of engagement decline relative to churn, limiting causal interpretation. It also does not incorporate transaction-level detail, customer service interactions, or competitor offers. Future work should apply predictive modeling (e.g., logistic regression or gradient boosting) to estimate individualized churn probability, incorporate time-series engagement trends, and validate the 'sticky customer' profile against post-intervention outcomes.

7. Conclusion

This project reframes customer churn from a behavioral and relationship-strength perspective. By focusing on engagement and product utilization rather than demographics or balance alone, it provides actionable, evidence-based insights for retention strategy design, product-bundling decisions, and the reduction of silent churn among premium customers.